Università della Svizzera Italiana

Faculty of Economics

Programming in Economics and Finance II

pyvar – A Python Package for Risk Management

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a.y. 2024/2025

Submission Date: June 13, 2025

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1 Introduction

Value-at-Risk (VaR) is a foundational tool in financial risk management. It measures the potential loss in value of a portfolio over a given time horizon, at a specified confidence level. VaR is widely used in practice—both by institutional investors for risk control and by regulators, such as the Basel Committee, to determine capital requirements. In most applications, it is computed using daily returns and short-term horizons (e.g., 1-day or 10-day), with confidence levels of 95% or 99%.

Despite its importance, many practitioners still rely on inefficient tools such as spreadsheets to compute VaR and related metrics. As programming becomes increasingly integrated into financial workflows, there is a growing need for accessible, reliable, and modern solutions for risk estimation.

We introduce `pyvar`, a Python package that automates the full VaR estimation pipeline—from portfolio creation to risk metrics, visualization, and backtesting. The package is designed for simplicity and speed, with a focus on equity portfolios and daily data. While its core features are tailored to equity, it also includes models capable of handling more complex portfolios involving options.

This document focuses on the theoretical background of the risk models implemented in `pyvar`. For a more practical overview of the package in action, including real-world use cases with historical data, we provide a series of Jupyter notebooks available in the project's GitHub repository: `https://github.com/Alessandro-Dodon/pyvar`.

2 Theoretical Foundation

This section outlines the mathematical foundations of our risk models. VaR estimation methods are traditionally grouped into three major categories: parametric, nonparametric, and simulation-based approaches. Each relies on different assumptions about the distribution of returns and offers distinct trade-offs between simplicity, flexibility, and computational cost.

In addition to this classical classification, we structure our models by the level of aggregation. We begin with univariate models, which describe the return dynamics of a single asset or a portfolio treated as an aggregated position. These models are often simpler and serve as the foundation for many practical applications.

We then introduce portfolio-level (multivariate) models, which consider the joint distribution of multiple asset returns. These approaches capture dependencies between assets and allow for a more accurate assessment of overall portfolio risk.

The section concludes with a presentation of the backtesting techniques used to evaluate the performance and reliability of the risk estimates produced by our models.

2.1 Basic VaR Models

Let $c \in (0, 1)$ denote the confidence level, and define $\alpha = 1 - c$ as the corresponding probability of a large loss. Value-at-Risk (VaR) at level α is defined as the worst loss L over a fixed horizon such that the probability of observing a larger loss is at most α:

$$\Pr(L > \mathrm{VaR}_\alpha) \leq \alpha.$$

Intuitively, this means we are confident that with probability c, losses will not exceed VaR_α over the given horizon.

By convention, VaR is expressed as a positive monetary amount. To simplify the notation—especially in univariate models—we omit the scaling by invested wealth W and use percentage-based notation instead. In the package, wealth is considered optional in univariate models, so percentage-based VaR is the default output. If specified, monetary VaR can be computed easily. In portfolio-level models, wealth is generally inferred from the parameters, and VaR is always returned in monetary units.

To simplify the calculations we assume that W is strictly positive and that returns follow a zero-mean process.

We define the loss variable as $\ell_t = -r_t$. We start with models that assume constant volatility over time. The non-parametric (historical) and the parametric normal VaR are computed as:

$$\mathrm{VaR}_\alpha^{\mathrm{hist}} = Q_\alpha(\ell_t), \qquad \mathrm{VaR}_\alpha^{\mathrm{normal}} = z_\alpha \sigma,$$

where $Q_\alpha(\ell_t)$ is the empirical α-quantile of losses, and $z_\alpha = \Phi^{-1}(1-\alpha)$ is the positive quantile of the standard normal distribution. For the Student-t VaR, the quantile is taken from the fitted Student-t distribution, estimated via maximum likelihood.

Normal VaR is rarely reliable for daily financial data, as it tends to underestimate extreme losses. In our empirical experiments, Student-t models offer significantly better tail fit and risk estimation. In practice, nonparametric (historical) methods are often preferred for their simplicity and robustness.

Expected Shortfall (ES), also known as Conditional VaR, is computed as:

$$\mathrm{ES}_\alpha^{\mathrm{historical}} = \mathbb{E}\left[\ell_t \mid \ell_t \geq \mathrm{VaR}_\alpha\right], \quad \mathrm{ES}_\alpha^{\mathrm{normal}} = \sigma \cdot \frac{\phi(z_\alpha)}{\alpha}, \quad \mathrm{ES}_\alpha^{t_\nu} = \sigma \cdot \frac{f_{t_\nu}(z_\alpha)}{\alpha} \cdot \frac{\nu + z_\alpha^2}{\nu - 1}$$

where ϕ is the standard normal density and f_{t_ν} is the density of the Student-t distribution with ν degrees of freedom.

The Cornish-Fisher expansion adjusts the normal quantile to account for higher-order moments of the return distribution — particularly skewness and kurtosis. This leads to a closed-form approximation of VaR:

$$\mathrm{VaR}_\alpha^{\mathrm{CF}} = -z^{\mathrm{CF}} \cdot \sigma,$$

where the adjusted quantile z^{CF} is given by:

$$z^{\mathrm{CF}} = z + \frac{1}{6}(z^2 - 1)s + \frac{1}{24}(z^3 - 3z)k - \frac{1}{36}(2z^3 - 5z)s^2,$$

with s and k representing the sample skewness and excess kurtosis, and z the standard normal quantile.

This expansion allows for asymmetry and fat tails in a parametric way, though it does not extend naturally to a consistent Expected Shortfall estimate. In practice, the Cornish-Fisher adjustment is only effective when deviations from normality are mild. It can become unstable in the presence of large skewness or kurtosis, leading to unreliable risk estimates.

The Hybrid or Exponentially Weighted Historical approach refines the historical approach by applying an exponentially decaying weighting scheme to past losses, giving more relevance to recent observations. The weights are defined as:

$$w_t = \alpha_0 \cdot \lambda^t, \quad \text{where } \alpha_0 = \frac{1-\lambda}{\lambda(1-\lambda^n)},$$

for $t = 0, 1, \ldots, n-1$, and $\lambda \in (0, 1)$ is a decay factor.

VaR and ES are computed from the weighted loss distribution as:

$$\mathrm{VaR}_\alpha^{\mathrm{hybrid}} = Q_\alpha^{\mathrm{w}}(\ell_t), \qquad \mathrm{ES}_\alpha^{\mathrm{hybrid}} = \frac{\sum_{t:\, \ell_t \geq \mathrm{VaR}_\alpha} w_t \ell_t}{\sum_{t:\, \ell_t \geq \mathrm{VaR}_\alpha} w_t}.$$

This exponentially weighted scheme is widely used in finance due to its ability to adapt quickly to changing volatility conditions. It significantly improves historical VaR estimates by mitigating the so-called ghost effect—where outdated extreme events unduly inflate risk estimates long after they are relevant.

2.2 Extreme Value Theory

Technically still a parametric model, EVT is more sophisticated than the methods discussed above. We implement it with the Peaks-Over-Threshold (POT) approach: once a high threshold u (e.g., the 99th-percentile loss) is chosen, a Generalized Pareto Distribution (GPD) is fitted to the excesses above u.

VaR and ES are then derived analytically from the fitted GPD parameters. This allows for more accurate estimation of rare, extreme losses. The formulas used are:

$$\mathrm{VaR}_\alpha = u + \frac{\beta}{\xi}\left[\left(\tfrac{n}{n_u}(1-c)\right)^{-\xi} - 1\right], \qquad \mathrm{ES}_\alpha \quad = \frac{\mathrm{VaR}_\alpha + \beta - \xi u}{1-\xi},$$

where ξ and β are the estimated shape and scale parameters, n is the sample size, and n_u is the number of exceedances.

However, the accuracy of this approach depends heavily on having a sufficiently large dataset. Fitting the tail of the distribution requires enough exceedances over the threshold to reliably estimate the shape and scale parameters. When used appropriately, EVT enables estimation of VaR and ES at extremely high confidence levels (e.g., 99.97% or 99.99%), which are nearly impossible to obtain with standard parametric or nonparametric methods.

In one of our example notebooks, we demonstrate how this method becomes particularly powerful when combined with simulation. Profit and loss scenarios can be generated using any of our simulation methods, which will be described in detail below. This setup ensures an abundance of tail data, allowing EVT to fit the tail parameters more effectively and yield robust estimates of extreme risk measures.

2.3 Volatility Models

We have implemented the most popular univariate volatility models.

The GARCH(1,1) model is specified as:

$$\sigma_t^2 = \omega + a\, r_{t-1}^2 + b\, \sigma_{t-1}^2, \qquad r_t = \sigma_t z_t,$$

where z_t are i.i.d. standardized innovations. VaR is then computed as:

$$\mathrm{VaR}_t = \sigma_t z_\alpha.$$

Here, z_α is the positive α-quantile from the empirical distribution of the standardized innovations.

Future volatility can be forecasted analytically using GARCH(1,1).

The τ-step ahead forecast and cumulative T-step forecast are:

$$\begin{aligned}
\mathbb{E}[\sigma_{t+\tau}^2] &= \mathrm{VL} + (a+b)^\tau (\sigma_t^2 - \mathrm{VL}), \\
\mathbb{E}[\sigma_{t,T}^2] &= \mathrm{VL}\left(T - 1 - \frac{(a+b)(1-(a+b)^{T-1})}{1-(a+b)}\right) + \sigma_t^2\, \frac{1-(a+b)^T}{1-(a+b)},
\end{aligned}$$

which naturally allow multi-period VaR estimation.

This approach provides an efficient and computationally fast way to forecast risk over multiple horizons. However, as we will discuss later, simulation-based methods offer an alternative that is more flexible and naturally extends to full portfolio-level VaR estimation.

We also support EGARCH (logs variance, capturing leverage), GJR-GARCH (adds an indicator for negative shocks), and APARCH (introduces a power term for flexible tail dynamics), all with maximum-likelihood estimation under Normal, Student-t, GED, or Skew-t innovations.

These volatility models are not just used as stand-alone tools: in later sections, we will show how they can be combined with factor models or simulation-based methods to produce more realistic, portfolio-wide risk models. These combinations are particularly effective for capturing cross-sectional risk dynamics in large portfolios.

For simpler benchmarks we include ARCH(p) and two rolling estimators, Moving Average (MA) and Exponentially Weighted Moving Average (EWMA):

$$\sigma_t^{2,\mathrm{ARCH}} = \omega + \sum_{i=1}^{p} \alpha_i\, r_{t-i}^2, \qquad \sigma_t^{2,\mathrm{MA}} = \frac{1}{n}\sum_{i=1}^{n} r_{t-i}^2, \qquad \sigma_t^{2,\mathrm{EWMA}} = \lambda \sigma_{t-1}^2 + (1-\lambda) r_{t-1}^2.$$

ES for volatility models becomes:

$$\mathrm{ES}_t = -\sigma_t\, \mathbb{E}[z_t \mid z_t < -z_\alpha].$$

All our volatility models use a semi-empirical approach for VaR computation: we first estimate the empirical distribution of standardized innovations z_t, then compute the required quantile z_α.

The importance of realistic conditional volatility modeling becomes especially clear in the backtesting stage. In particular, without accounting for time-varying volatility, VaR estimates systematically fail to satisfy the independence test—a critical requirement for the validity of risk models. This highlights why volatility models are a core component of modern risk estimation pipelines.

2.4 Time-Varying Correlation Models

We now introduce models designed to capture time-varying correlations at the portfolio level, extending beyond single assets. Specifically, we implement four approaches: moving average (MA), exponentially weighted moving average (EWMA), rolling principal component analysis (PCA), and Ledoit-Wolf (LW) shrinkage. These methods aim to provide more accurate and stable estimates of the conditional covariance matrix Σ_t, which governs portfolio-level risk.

Let $x_t = (x_{1,t}, \ldots, x_{N,t})^\top$ denote the vector of monetary positions, and let $w_t = x_t / \sum_{i=1}^N x_{i,t}$ be the corresponding vector of portfolio weights. We allow monetary positions in an asset to be negative, representing short positions. However, we assume the total portfolio value is strictly positive at all times.

Each method provides a different way to estimate the conditional covariance matrix. In the MA and EWMA models, Σ_t is estimated using a rolling sample or an exponentially weighted average of past return covariances, respectively.

In the PCA model, we compute a rolling sample covariance matrix and approximate it by retaining only the top eigencomponents, forming a low-rank matrix $\Sigma_t^{\mathrm{PCA}} = V_t \Lambda_t V_t^\top$, where V_t contains the leading eigenvectors and Λ_t the corresponding eigenvalues.

For Ledoit-Wolf shrinkage, we estimate the covariance over a rolling window and compute a regularized matrix $\Sigma_t^{\mathrm{LW}} = \delta_t F + (1 - \delta_t) S_t$, where S_t is the sample covariance matrix, F is a structured target (e.g., identity scaled by average variance), and $\delta_t \in [0, 1]$ is the shrinkage intensity estimated from the data.

In particular, the PCA and Ledoit–Wolf methods are used to de-noise the covariance estimate, which is especially important when dealing with large portfolios where the sample covariance matrix becomes unstable or ill-conditioned.

In all cases, the 1-day VaR is computed as:

$$\mathrm{VaR}_t = z_\alpha \cdot \sqrt{w_t^\top \Sigma_t w_t},$$

where Σ_t is the estimated variance-covariance matrix, obtained using one of the four methods described above: MA, EWMA, PCA, or Ledoit–Wolf shrinkage.

These models adapt naturally to both parametric and semi-empirical frameworks and assume a buy-and-hold strategy for portfolio positions. Expected Shortfall is computed using either the closed-form formula under normality or an empirical tail average of standardized residuals.

With all these models, the portfolio-level volatility can be directly computed at each time step, enabling efficient real-time risk tracking. This is demonstrated in the notebook examples, where we show practical applications of these techniques on large portfolios.

2.5 Factor Models

We implement the Sharpe model and Fama-French three factor models to estimate portfolio VaR and ES. These models use asset betas with respect to systematic risk factors, allowing us to infer portfolio-level risk from exposures to common sources of variation.

This approach is commonly known as risk mapping, as it projects the portfolio's risk onto a small set of underlying factors.

For the Sharpe Model, we assume that each asset's return r_i follows the linear factor structure:

$$r_i = \alpha_i + \beta_i r_m + \varepsilon_i,$$

where r_m is the return on the market portfolio, β_i is the asset's sensitivity to the market factor, α_i is the asset's abnormal return, and ε_i is a zero-mean idiosyncratic shock uncorrelated with r_m and other residuals. The market return r_m has variance σ_m^2, and $\mathrm{Var}(\varepsilon_i) = \sigma_{\varepsilon_i}^2$. We support both a simplified setting with constant market volatility and a dynamic version where σ_m^2 is estimated via a GARCH model. Portfolio weights are assumed to be perfectly constant.

The total portfolio variance is:

$$\sigma_p^2 = \left(\sum_{i=1}^{N} w_i \beta_i\right)^2 \sigma_m^2 + \sum_{i=1}^{N} w_i^2 \sigma_{\varepsilon_i}^2.$$

VaR and ES are computed as follows under the normality assumption:

$$\mathrm{VaR}_\alpha = z_\alpha \cdot \sigma_p, \qquad \mathrm{ES}_\alpha = \sigma_p \cdot \frac{\phi(z_\alpha)}{\alpha}.$$

We also extend the previous model by considering three risk factors: market excess return (Mkt-RF), size (SMB), and value (HML). Each asset's excess return is modeled as:

$$r_i - r_f = \alpha_i + \beta_{i,1} \cdot \text{Mkt-RF} + \beta_{i,2} \cdot \text{SMB} + \beta_{i,3} \cdot \text{HML} + \varepsilon_i,$$

where r_f is the risk-free rate, and ε_i is the idiosyncratic error. We estimate factor loadings $\beta_{i,k}$ through OLS regression.

Let B be the $N \times 3$ matrix of estimated factor loadings, Σ_f the 3×3 sample covariance matrix of the factors, and Σ_ε the diagonal matrix of residual variances $\sigma_{\varepsilon_i}^2$. The total portfolio variance is:

$$\sigma_p^2 = w^\top (B \Sigma_f B^\top + \Sigma_\varepsilon) w.$$

Once the portfolio variance is known, we compute VaR and ES as we did for the single factor model.

2.6 Analytic VaR

We compute several Analytic VaR measures using the previously defined monetary position vector $x_t = (x_{1,t}, \ldots, x_{N,t})^\top$ and the covariance matrix Σ.

The quantile z_α corresponds to the desired tail probability level and is always derived from the standard normal distribution, as the underlying assumption is that asset returns are normally distributed.

The asset-normal parametric portfolio VaR is defined as:

$$\mathrm{VaR}_t = z_\alpha \cdot \sqrt{x_t^\top \Sigma x_t} \cdot \sqrt{h}.$$

Here, h denotes the holding period, under the assumption of independent and identically distributed returns.

Assuming all asset returns are perfectly correlated ($\rho = 1$) and there are no short positions, the undiversified portfolio VaR is simply the sum of the individual asset VaRs:

$$\mathrm{UVaR}_t = \sum_{i=1}^{N} \mathrm{VaR}_{i,t}.$$

The marginal VaR, representing the sensitivity of portfolio VaR to a small change in position $x_{i,t}$, is given by:

$$\Delta\mathrm{VaR}_{i,t} = \mathrm{VaR}_t \cdot \frac{(\Sigma x_t)_i}{x_t^\top \Sigma x_t}.$$

It measures the change in total VaR from adding one extra unit of asset i.

The component VaR is:

$$\mathrm{CVaR}_{i,t} = x_{i,t} \cdot \Delta\mathrm{VaR}_{i,t},$$

which captures the absolute contribution of asset i to total portfolio VaR.

The relative component VaR is:

$$\mathrm{RCVaR}_{i,t} = \frac{\mathrm{CVaR}_{i,t}}{\mathrm{VaR}_t},$$

interpreted as the percentage share of total VaR attributable to asset i.

For a vector a representing changes in the portfolio allocation, the incremental VaR is:

$$\mathrm{IVaR}_t = \Delta\mathrm{VaR}_t^\top \cdot a,$$

which estimates the change in total VaR resulting from shifting the portfolio by a.

Each of these VaR metrics can be complemented with a corresponding Expected Shortfall (ES), computed under the normality assumption.

2.7 Options

Delta-Normal approximations provide fast and analytically tractable methods for estimating the VaR of portfolios containing European options. These methods rely on the Black-Scholes model, which not only underpins the delta-based risk approximation used here but also serves as a foundation for the simulation-based techniques discussed in the next section.

This approach is another form of risk mapping: it translates the non-linear risk of options into equivalent linear exposures (deltas) to the underlying assets, allowing standard portfolio VaR techniques to be applied.

The Black-Scholes model gives the fair value of a European call or put option as:

$$c_t = N(d_1)S_t - N(d_2)Ke^{-r(T-t)}, \qquad p_t = N(-d_2)Ke^{-r(T-t)} - N(-d_1)S_t,$$

where

$$d_1 = \frac{\ln(S_t/K) + \left(r + \frac{1}{2}\sigma^2\right)(T-t)}{\sigma\sqrt{T-t}}, \qquad d_2 = d_1 - \sigma\sqrt{T-t}.$$

Here, S_t is the spot price, K the strike price, $T-t$ the time to maturity in years, r the continuously compounded risk-free rate, σ the annualized volatility of returns, and $N(\cdot)$ the standard normal cumulative distribution.

The option's delta, which measures sensitivity to the underlying asset, is given by $\Delta = N(d_1)$ for a call and $\Delta = N(d_1) - 1$ for a put. Using this, a first-order approximation of the VaR_α for a single option position with q contracts, each on N units of the underlying asset, is:

$$\mathrm{VaR}_\alpha = z_\alpha \cdot |\Delta| \cdot S \cdot \sigma \cdot q \cdot N \cdot \sqrt{h},$$

where z_α is the positive quantile of the standard normal distribution corresponding to the confidence level.

For a portfolio of options on multiple underlying assets, we compute the dollar delta exposure for each asset as $x_i = \Delta_i \cdot S_i \cdot q_i \cdot N_i$, and estimate VaR_α using the variance-covariance matrix Σ of asset returns:

$$\mathrm{VaR}_\alpha = z_\alpha \cdot \sqrt{\mathbf{x}^\top \Sigma \mathbf{x}} \cdot \sqrt{h}.$$

These formulas provide a fast approximation of market risk, assuming that portfolio values are linearly sensitive to underlying prices and that asset returns are normally distributed. However, when portfolios contain complex or highly nonlinear positions, these assumptions break down. In such cases, simulation-based approaches—which will be introduced next—offer a more flexible and accurate alternative.

2.8 Simulation Methods

Simulation methods, also known as full valuation approaches, are powerful tools for estimating VaR and ES—especially for portfolios containing non-linear instruments such as options. In our package, these methods enable risk estimation for portfolios that include both equity and options, as well as for equity-only or options-only portfolios. This flexibility is demonstrated in the notebook examples.

By default, all simulation methods estimate one-day ahead VaR, ES, and simulated profit and loss (P&L). Some multi-day extensions are also supported, enabling risk forecasting over longer horizons.

We begin with the one-day Monte Carlo approach under the assumption of multivariate normality. Future returns are simulated as:

$$\mu = \mathbb{E}[r_t], \quad \Sigma = \mathrm{Cov}(r_t), \quad L = \mathrm{Cholesky}(\Sigma),$$

$$r^{(i)} = \mu + L z^{(i)}, \quad z^{(i)} \sim \mathcal{N}(0, I),$$

$$S^{(i)} = S_0 \circ \exp(r^{(i)}),$$

where $\circ$ denotes element-wise multiplication. Portfolio profit and loss is calculated as:

$$\Delta P^{(i)} = w^\top (S^{(i)} - S_0) + \sum_{j=1}^{N_{\mathrm{opt}}} q_j \left(C(S_j^{(i)}, \tau_j') - C_{j,0} \right),$$

where q_j is the number of option contracts, $C(\cdot)$ is the Black-Scholes option price, and $\tau_j' = \max(T_j - \frac{1}{252}, 0)$ is the adjusted time to maturity.

We define simulated portfolio losses as $\ell^{(i)} = -\Delta P^{(i)}$. From the empirical distribution $\{\ell^{(i)}\}_{i=1}^N$, where N is the number of simulations, we compute:

$$\mathrm{VaR}_\alpha = Q_\alpha(\ell), \qquad \mathrm{ES}_\alpha = \mathbb{E}[\ell \mid \ell \geq \mathrm{VaR}_\alpha].$$

This method is extended to multiple days (equity-only), where asset paths are iteratively simulated and used to compute terminal portfolio losses. These multi-step simulations serve as a forward-looking, non-analytical forecast of risk.

To better capture return asymmetry and regime switching, we implement a Gaussian Mixture Model (GMM) extension. In the multivariate case:

$$r^{(i)} \sim \sum_{k=1}^{K} \pi_k \cdot \mathcal{N}(\mu_k, \Sigma_k),$$

where K is the number of Gaussian components (typically between 2 and 4), and π_k are the mixture weights. Prices are propagated like before.

We also support a univariate GMM version, which fits a mixture to log-returns of a single asset. As with Monte Carlo, both univariate and multivariate GMMs can be extended to multiday horizons, although only for equity portfolios.

We also implement univariate GARCH-based simulations. A GARCH(1,1) model is fitted to historical log-returns, and future volatility paths are simulated under zero drift ($\mu = 0$) to avoid artificial upward trending in asset paths. Two innovation types are supported: standard normal shocks and standardized Student-t shocks for heavier tails. Prices are again obtained using the same exponential propagation.

As a non-parametric alternative, we incorporate a suite of Historical Simulation methods with varying complexity and adaptability: standard Historical Simulation, a bootstrap-enhanced variant, Weighted Historical Simulation that emphasizes recent data, and Filtered Historical Simulation incorporating conditional volatility.

A key advantage of historical methods is that they completely bypass the need to explicitly model or estimate the portfolio's variance-covariance matrix. Instead, they implicitly preserve the dependence structure among assets by reusing actual past return vectors—i.e., the returns observed on the same historical day—when generating scenarios.

In the standard version, historical log-returns are applied directly to current prices using the same exponential propagation as in the parametric models. The bootstrap variant resamples past return vectors with replacement, which can improve robustness in small-sample settings and help reduce overfitting to specific past events.

The Weighted Historical Simulation variant improves responsiveness to recent market behavior by applying exponentially decaying weights.

This matches the approach used in hybrid VaR and ES, applying exponential weights to emphasize recent losses.

Finally, Filtered Historical Simulation combines GARCH modeling with empirical innovations. A GARCH model is fitted to each asset, and standardized residuals z_t are either directly used or resampled, then rescaled using the most recent conditional volatility $\hat{\sigma}_T$:

$$r^{(i)} = \hat{\sigma}_T \cdot z^{(i)},$$

with returns propagated to prices as before. This approach captures time-varying volatility while remaining non-parametric in distributional assumptions.

Across all methods, except for the WHS case, Expected Shortfall is computed as:

$$\mathrm{ES}_\alpha = \mathbb{E}[\ell \mid \ell \geq \mathrm{VaR}_\alpha].$$

We recognize the following limitations of our simulation framework. All methods assume static portfolio weights and do not incorporate stochastic volatility models such as Heston, or stochastic interest rate models like Vasicek or CIR. Instead, our setup is intentionally simplified and built around the Black-Scholes framework, as introduced earlier. This choice enables efficient full-valuation risk estimation while keeping the implementation practical and focused. For computational reasons, no backtesting functionality is included within the simulation methods.

2.9 Backtesting

Backtesting assesses whether the observed losses are consistent with the VaR predictions by checking the frequency and structure of violations (exceptions). The objective is to test whether the model is correctly calibrated.

We implement three standard tests. The Kupiec test (unconditional coverage) evaluates if the number of observed exceptions N over T days matches the expected number under the model. If p is the failure probability, then under the null N follows a binomial distribution. The likelihood ratio test statistic is:

$$\mathrm{LR}_{\mathrm{uc}} = -2\left\{\ln\left[(1-p)^{T-N}p^N\right] - \ln\left[\left(1-\frac{N}{T}\right)^{T-N}\left(\frac{N}{T}\right)^N\right]\right\}, \quad \mathrm{LR}_{\mathrm{uc}} \sim \chi_1^2.$$

The Christoffersen test (conditional coverage) tests the independence of exceptions by modeling their dynamics as a first-order Markov chain. It compares the likelihoods of transition counts under the null (independence) and alternative. The statistic is:

$$\mathrm{LR}_{\mathrm{c}} = -2(\ln L_0 - \ln L_1), \quad \mathrm{LR}_{\mathrm{c}} \sim \chi_1^2,$$

where L_0 is the likelihood under independence and L_1 under the alternative.

The joint test combines both:

$$\mathrm{LR} = \mathrm{LR}_{\mathrm{uc}} + \mathrm{LR}_{\mathrm{c}} \sim \chi_2^2.$$

Rejecting the null in any of the tests suggests that the VaR model is either miscalibrated (too many or too few violations) or fails to capture time dependence in risk (e.g., volatility clustering).

Our back-testing framework covers every method except the simulation models.

Beyond statistical calibration, backtesting intuitively helps answer the question: ”Which model is the best?” A good model is not only one that aligns with financial theory and empirical facts—such as volatility clustering, conditional heteroskedasticity, and fat tails—but also one that consistently passes validation tests. However, care must be taken to avoid overfitting to a particular dataset or test. For this reason, stress testing is also commonly used in practice as a complementary tool, although it is not automatically supported in the current version of the package.

3 Results and Applications

Virtually all applications and results of the `pyvar` package can be found in the `examples` folder of the GitHub repository. A series of Jupyter notebooks illustrate typical use cases, all following a consistent setup: the creation of a simple portfolio followed by the use of the package’s functions to calculate various risk metrics and visualize them effectively. Each notebook includes additional insights and brief theoretical refreshers to support understanding and contextualize the results.

In addition to the notebooks, we report here selected example outputs and backtesting results. For univariate models, we analyze an investment of 100,000 USD in the global equity ETF VT (Vanguard) from 1 January 2000 to 1 January 2025. Using the parametric normal model, the estimated 1-day monetary VaR at the 99% confidence level is 3,065.52 USD. However, the model fails backtesting on all fronts: *Total Violations*: 80, *Violation Rate*: 1.92%, and it fails both the joint and independence tests. A plot of the backtesting violations is shown in Figure 1. In contrast, the same experiment using a GJR-GARCH(1,1) model with skew-t innovations produces a more realistic, conditional VaR estimate, passing all backtesting tests with *Total Violations*: 42 and *Violation Rate*: 1.01%. The corresponding backtesting plot, shown in Figure 2, focuses on a high-volatility subset of the full sample period.

We also report results for a globally diversified ETF and stock portfolio constructed from long and short positions in assets including VT, SPY, VGK, EWJ, VWO, AAPL, JNJ, PG, UNH, NVS, and a short position in XLI. The portfolio is held over the period from 1 January 2023 to 1 January 2025, and the total equity value of 103,642.05 CHF refers to the final value of the portfolio's equity component on the last day. Using a single-factor model (with VT as the market benchmark) and APARCH(1,1) for conditional volatility, the model yields *Total Violations*: 6 and *Violation Rate*: 1.20%, with all tests passed. This setup computes risk in USD, and the final portfolio value is 114,744.75 USD. Using the Fama–French three-factor model over the same period gives *Total Violations*: 5 and *Violation Rate*: 1.00%, again passing all tests. Figure 3 shows the estimated portfolio volatility under both factor models.

For time-varying correlation models, we backtest four approaches: moving average with a 60-day window yields *Total Violations*: 5 and *Violation Rate*: 1.13%; EWMA with $\lambda = 0.94$ gives *Total Violations*: 5 and *Violation Rate*: 1.00%; rolling PCA with three components and Ledoit–Wolf shrinkage (both with a 20-day window) each yield *Total Violations*: 5 and *Violation Rate*: 1.04%. All models pass the joint, unconditional, and independence tests. Empirical innovations are used throughout. Figure 4 displays the portfolio volatility from the four correlation models.

For simulation-based methods, we augment the previous portfolio by including three European-style options: a long call on AAPL (3,000 contracts), a long put on XLI (5,000 contracts), and a long call on VT (2,500 contracts), all priced using the Black–Scholes formula at time 0. The respective values are 8,340,662.05 CHF, 902,882.06 CHF, and 1,781,738.50 CHF, resulting in a total option value of 11,025,282.61 CHF and a total portfolio value of 11,128,924.66 CHF. The following 1-day VaR estimates at the 99% confidence level are obtained: Monte Carlo simulation yields 4,412,141.49 CHF; bootstrapped historical gives 4,319,563.44 CHF; Filtered Historical Simulation (FHS) produces 5,768,723.40 CHF; Weighted Historical Simulation (WHS) gives 5,584,225.70 CHF; and GMM-based Monte Carlo results in 4,775,783.75 CHF. Figure 5 shows the KDE plots of simulated profit and loss (P&L) distributions and the corresponding VaR estimates for each simulation method.

4 References

The following textbooks and lecture notes constitute the main references for this project:

- John C. Hull, *Risk Management and Financial Institutions*, 2023, John Wiley & Sons.
- Peter Christoffersen, *Elements of Financial Risk Management* (2nd Edition), Academic Press, 2011.
- Philippe Jorion, *Value at Risk*, McGraw-Hill, 2006.
- Andrea Resti and Andrea Sironi, *Risk Management and Shareholders' Value in Banking*, John Wiley & Sons, 2007.
- John C. Hull, *Options, Futures, and Other Derivatives*, latest edition, Pearson.
- Kevin Sheppard, *Financial Econometrics Notes*, University of Oxford, February 2, 2021.

Additional insights and context were drawn from the following MSc in Quantitative Finance courses at Università della Svizzera Italiana, Lugano:

- *Risk Management*, taught by Prof. A. Plazzi
- *Derivatives and Financial Econometrics*, taught by Prof. L. Mancini

We also acknowledge the valuable content and feedback from:

- Francesco Violante, *Market Risk Management*, IÉSEG School of Management, Paris – kindly suggested by Andrea Mercier and Farah Saoudi, who also provided insightful comments on several modules of the package.

5 Appendix

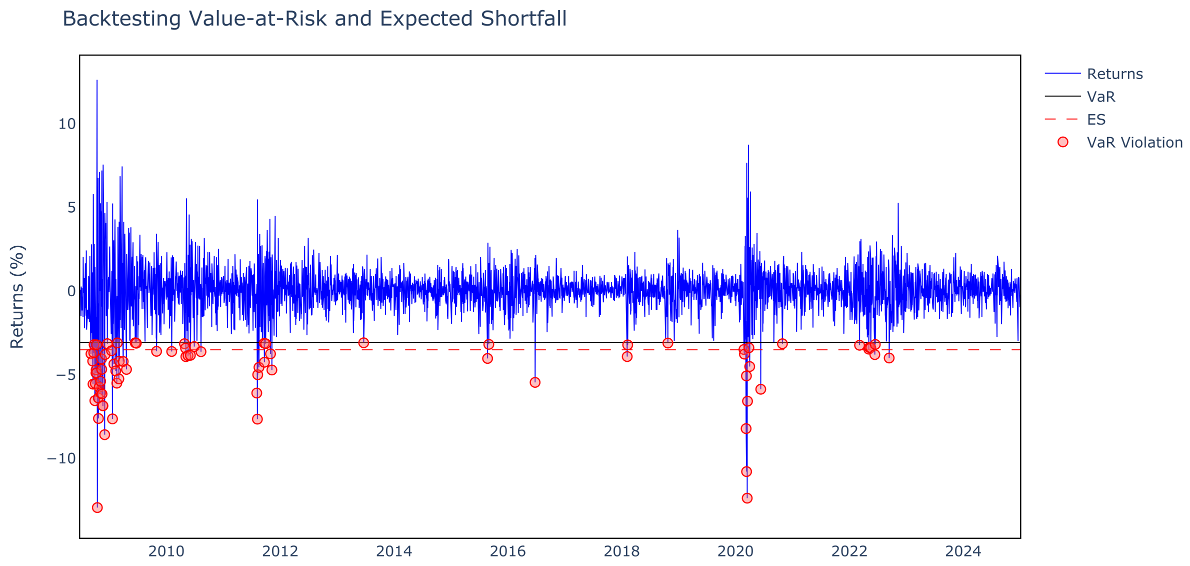

Figure 1: Backtesting plot for the normal model applied to VT (2000–2025).

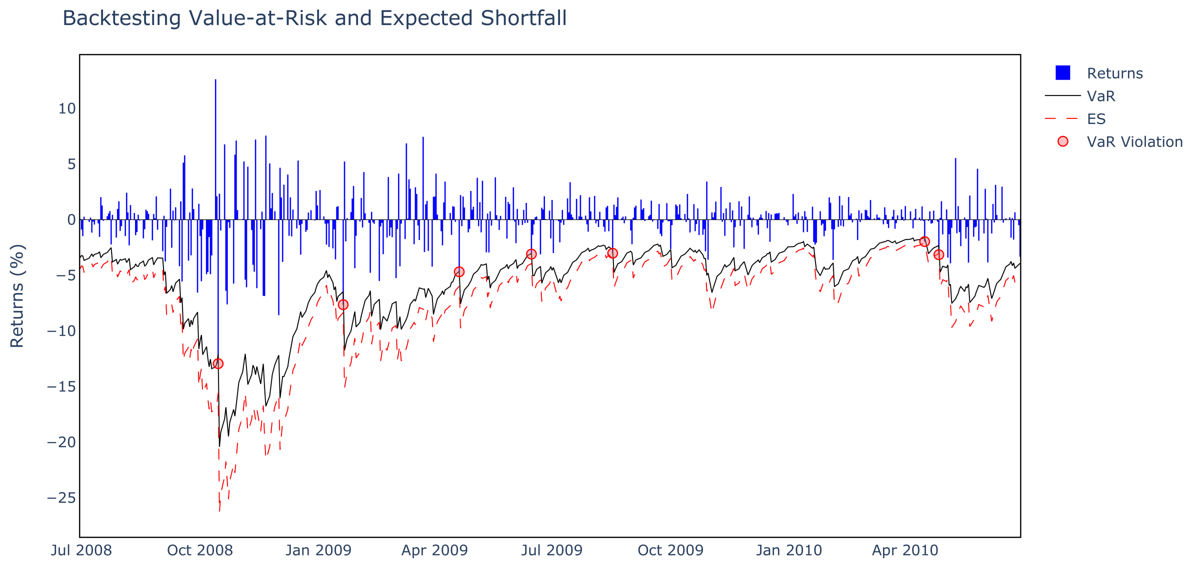

Figure 2: Backtesting plot for the GJR-GARCH(1,1) model on a high-volatility subset.

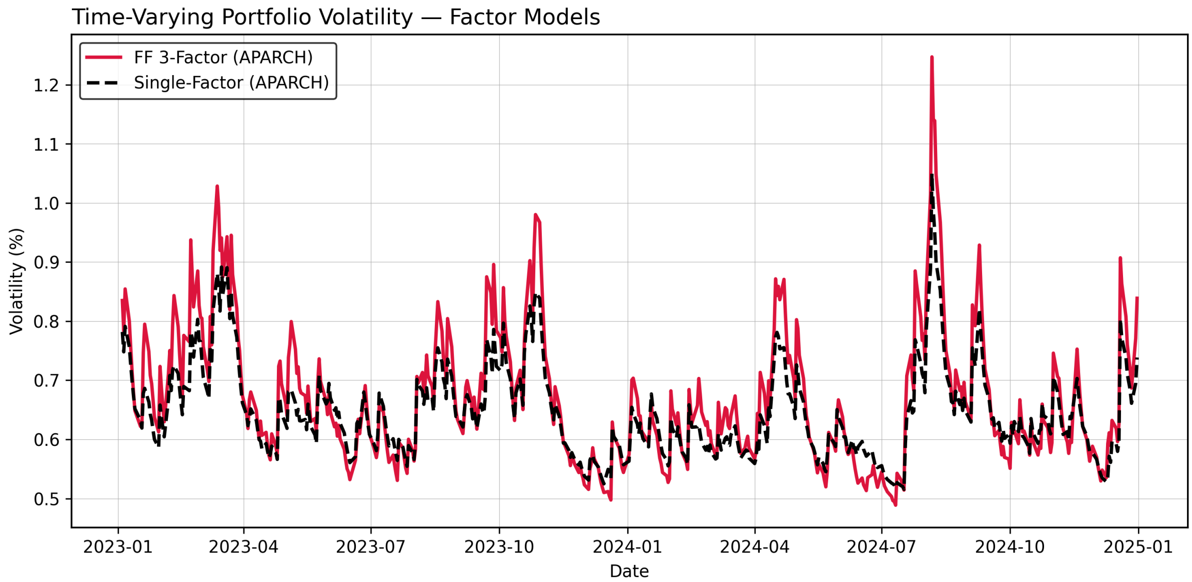

Figure 3: Estimated portfolio volatility under the single-factor and Fama–French three-factor models.

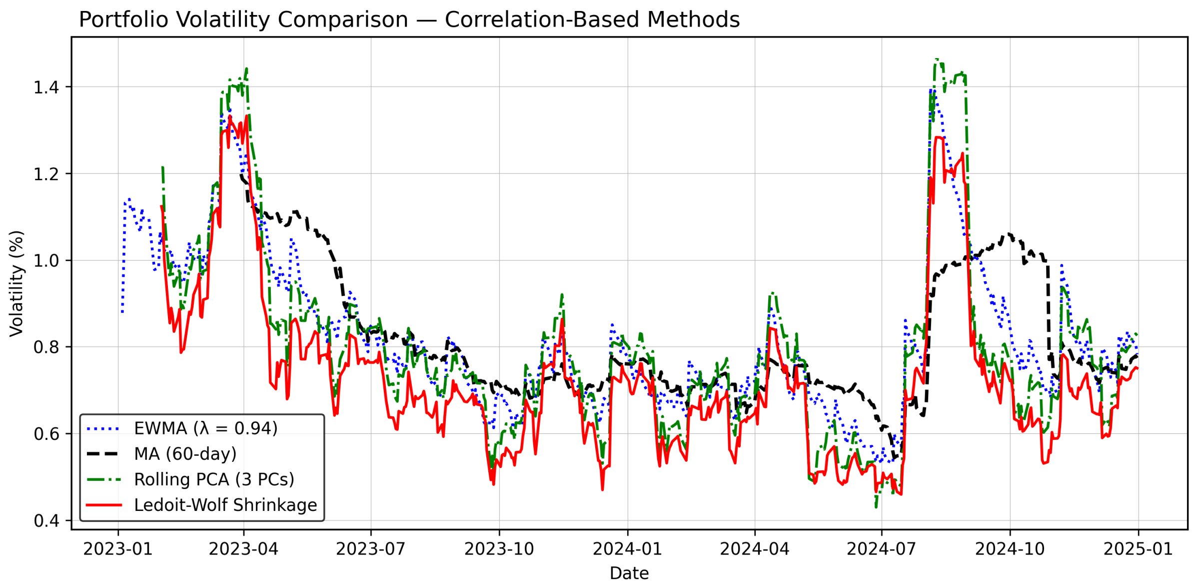

Figure 4: Portfolio volatility estimated using MA, EWMA, PCA, and Ledoit–Wolf correlation models.

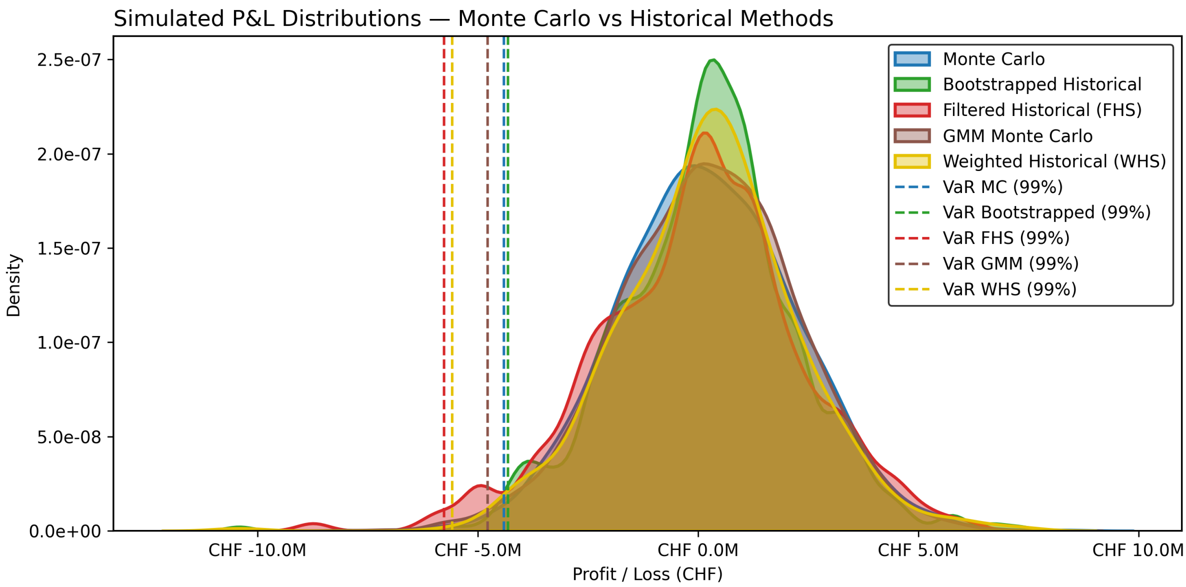

Figure 5: KDEs of simulated profit and loss with 1-day VaR for each simulation method.